

Hee Won Lee

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SUMMARY

Cloud infrastructure and software engineer with deep expertise in designing and optimizing distributed systems. At Samsung, led the Cloud Platform team in building and operating infrastructure services across cloud and on-premises datacenters. Developed Kubernetes platforms, DevOps pipelines, observability stacks, infrastructure as code, and GenAI services. At AT&T, developed Kubernetes-orchestrated Ceph storage and Redis in-memory databases. Also, designed and implemented network systems for OpenStack clusters. Holds a Ph.D. in Computer Science with a focus on virtualization (QEMU-KVM), software-defined networking (SDN), and systems performance optimization.

EXPERIENCE

Samsung Electronics, Hwaseong, South Korea

Principal Engineer

Dec 2020 – Feb 2024

- Hands-on Software Development (Kubernetes, Ceph, GitOps, Observability, IaC, and GenAI)
 - Developed a highly-available cluster service using open-source Kubernetes by deploying control plane components (kube-apiserver, etcd, etc) behind a load balancer in a dedicated management cluster
 - Provisioned and managed Ceph storage clusters on Kubernetes using the Rook operator
 - Developed Kubernetes Custom Resources and controllers using the Operator SDK (w/ Golang)
 - Developed GitHub Actions pipelines (w/ SonarQube) to deploy apps to Kubernetes clusters (w/ Argo CD)
 - Developed observability stacks for K8s services w/ Jaeger, Prometheus, and Grafana using OpenTelemetry (w/ gRPC)
 - Developed telemetry systems for VM services with Telegraf, InfluxDB, and Grafana
 - Developed infrastructure as code for Azure and AWS using ARM template/Bicep and Terraform
 - Developed an LLM-powered chat service w/ retrieval-augmented generation (RAG) using the LangChain framework
- Technical Leadership
 - As Head of Group for Cloud Platform (reporting to the SVP), led eight projects, including:
 1. Software development – HA Kubernetes service, HPC batch scheduler, and cloud cost monitoring system;
 2. System engineering – VMware VM/VDI services, OpenStack VM service, and HA Active Directory;
 3. Public cloud architecture – Azure/AWS governance framework (landing zone) and AWS Outposts evaluation.
 - Delivered large-scale infrastructure services – VM (5K+ instances), VDI (10K+ instances), HPC batch scheduler (10K+ servers), HA Kubernetes (500+ servers), and HA Active Directory (60K+ users)
 - Managed 24/7 VM/VDI services with a 99.95% SLO (< 4 hours downtime/year) / zero downtime for Active Directory
 - Led team growth from 30 to over 60 engineers (SW/DevOps Engineers and Cloud Architects) within three years
 - Defined MBOs and developed KPIs for team projects (\$30M+/year budget), and achieved 100% goal completion
 - Developed data migration strategy for Microsoft SharePoint, applying Double Key Encryption for confidential data
 - Developed Kubernetes security best practices - image scan w/ Trivy, container runtime w/ non-root user, secret w/ Azure Key Vault, and ingress w/ TLS/mTLS

AT&T Labs Research, Bedminster, New Jersey, USA

Principal Inventive Scientist

Mar 2015 – Jul 2020

- Kubernetes & Observability
 - Designed and implemented containerized Ceph with Kubernetes helm charts (with an one-osd-per-pod approach)
 - Developed real-time monitoring systems for containerized Ceph with Telegraf, InfluxDB, and Grafana
 - Developed best recovery practices from various failure scenarios for containerized Ceph storage with Kubernetes
 - Designed and implemented Redis High Availability with Kubernetes in CI/CD environment (Gerrit and Jenkins)
 - Integrated Redis HA helm charts into Radio Access Network Intelligent Controller release 4 (field trial version)
- Storage Systems
 - Designed and implemented I/O traffic protection mechanisms for cloud storage
 - Designed and implemented an iSCSI block storage backend with a commodity SSDs for OpenStack clusters
 - Designed and implemented the backend of OpenStack Cinder for dynamic configuration and QoS support
 - Evaluated cache performance across host and VM page cache, dm-cache, and Ceph RBD cache
 - Evaluated the latency of Redis with Intel Optane Persistent Memory using the *memtier_benchmark* tool
 - Evaluated and compared the performance of *Redis* vs *Aerospike* vs *Cassandra* using the YCSB benchmark tool
 - Designed and implemented high-performance VMs by applying CPU pinning and SR-IOV PCI passthrough
- Network Systems
 - Designed and implemented a system that dynamically controls network bandwidth and latency between OpenStack Neutron deployments with Ansible (using Open vSwitch and Tc/NetEm)
 - Set up a testbed for RoCE-based lossless networks (using PFC-capable Mellanox Spectrum Switch and ConnectX-5 NICs) and evaluated the performance of NVMe over Fabric for zero CPU usage on the target host machine
 - Evaluating the performance of Apache Traffic Server with Intel Optane Persistent Memory
 - Analyzed cost benefits by applying Intel Optane Persistent Memory to AT&T's Content Delivery Network (CDN)

North Carolina State University, Raleigh, North Carolina, USA

- Research Assistant Jan 2010 – Dec 2014
 - Designed and implemented a virtualization platform for evaluating distributed applications running on diverse OSs (Linux, FreeBSD, Windows, JunOS)
 - Redesigned and customized the QEMU-KVM hypervisor to synchronize distributed VMs with virtual time

Korea Telecom, Daejeon, South Korea

- Software Engineer Jan 2002 – Jul 2009
 - Capacity Planning for KT Backbone Network and IPTV Services
 - Built a virtual network environment of KT backbone networks using OPNET simulation tool
 - Performed link capacity planning for IPTV services
 - VoIP Network Management System
 - Conducted requirement management and then designed architecture of VoIP-NMS
 - Deployed VoIP-NMS to production
 - IP Core Network Management System
 - Designed and implemented a system component that collects network equipment information from MIB (Management Information Base) using SNMP (Small Network Management Protocol)
 - Developed SQL scripts for Oracle database
 - Developed a system component that exchanges information with Service Assurance System using XML-RPC
 - Set up an MPLS testbed using Cisco Routers 7204/7500, Juniper Router M5 and Ixia Traffic Generator

CERTIFICATIONS **Microsoft Azure Solutions Architect Expert (AZ-305)** / Azure Administrator Associate (AZ-104)

Microsoft Cybersecurity Architect Expert (SC-100) / Azure Security Engineer Associate (AZ-500)

EDUCATION **North Carolina State University, Raleigh, North Carolina, USA**

- Ph.D. in Computer Science Aug 2009 – May 2015
 - Concentration: Virtualization, Networked Systems, Distributed Systems

Carnegie Mellon University, Pittsburgh, Pennsylvania, USA

- M.E. in Software Engineering Aug 2004 – Aug 2005

**SOFTWARE
DEVELOPMENT
EXPERIENCE**

CPU & Memory

- CPU/Memory Control [Python]
 - Designed and implemented a CPU/Memory control algorithm that enforces CPU/Memory ceiling and allocates a relative share of CPU time using Linux Control Groups (Cgroups)
- Multi-Threading [C++]
 - Created a thread to search text file paths and multiple worker threads to count words using Boost Filesystem and Thread libraries.
 - Implemented multiple producer, multiple consumer thread-safe queue and map
- Prioritized Preemptive Schedulers [C]
 - Implemented a prioritized preemptive scheduler using POSIX thread library
 - Designed and implemented a priority-based scheduling algorithm using XINU kernel (a small UNIX OS)
- Demand Paging [C]
 - Implemented a demand paging system that allows for more address space than physically available one (XINU)

Networking & IO

- I/O Traffic Protection [Python]
 - Designed and implemented an I/O bandwidth reservation algorithm that guarantees the minimum I/O bandwidth of a virtual host connected to a shared storage volume through iSCSI
 - Designed a NUMA-aware CPU pinning algorithm that protects I/O traffic from CPU interference
 - Evaluated the system performance for diverse types of I/O pattern using FIO and Filebench
- Network Link Emulation [C++, Python]
 - Interconnected distributed VMs through virtual links using Tc/NetEm with adaptive time dilation
- Software-Defined Networking [Python]
 - Used OpenFlow POX controller and Open vSwitch to migrate VirtualBox-based virtual networks
- Routing Algorithms [Java]
 - Implemented modified Dijkstra's algorithm with negative weights
 - Implemented vertex/edge disjoint paths and elementary circuits search algorithms
 - Implemented CPLEX code with Integer Linear Programming for Routing and Wavelength Assignment

- **Queueing Systems [Java]**
 - Implemented discrete event simulation models for queueing systems (M/M/1, M/M/m, M/G/m)
- **Wireless Networking [C++]**
 - Designed and implemented an OMNeT++ model for IVG (Inter-Vehicle Geocast) routing protocol in vehicle-to-vehicle ad-hoc network simulation
 - Connected KVM-based virtual nodes through a wireless simulation model of OMNeT++
- **Peer-to-Peer File Sharing System [Java, Apache Mina, Apache FtpServer, JavaDB]**
 - Mapped a file signature (created by encoding text document with m-bit string) onto a hash space

Virtualization

- **Hypervisor (QEMU-KVM) [C++]**
 - Designed and implemented a spinlock on shared memory to synchronize distributed VMs (processes created by QEMU-KVM hypervisor) and simulation nodes (processes created by NS-3) with virtual time
 - Implemented a virtual time synchronization daemon with TCP/UDP sockets using Boost Asio Library
- **High-Performance Emulation System [C++, Python]**
 - Emulated higher performance with a time dilation technique using unmodified OSs (Linux, FreeBSD, Windows, and Junos) and real application workloads (VLC media player)
 - Modified the QEMU-KVM hypervisor for virtual time
- **Hybrid Simulation and Emulation System [C++, Python]**
 - Built an integrated simulation (NS-3) and emulation (QEMU-KVM) framework in a distributed environment

PUBLICATIONS

DISTRIBUTED STORAGE SYSTEMS

- **MIST: Mitigating Host-Side Interference for Storage Traffic in Virtualized Data Centers.**
Hee Won Lee and Moo-ryong Ra
 In *IEEE 9th International Conference on Cloud Computing (CLOUD 2016)*.
- **Fighting with Unknowns: Estimating the Performance of Scalable Distributed Storage Systems with Minimal Measurement Data.**
 Moo-ryong Ra and Hee Won Lee.
 In *IEEE 35th Symposium on Mass Storage Systems and Technologies (MSST 2019)*.
- **eMRC: Efficient Miss Ratio Approximation for Multi-Tier Caching.**
 Zhang Liu, Hee Won Lee, Yu Xiang, Dirk Grunwald, and Sangtae Ha.
 In *USENIX 19th Conference on File and Storage Technologies (FAST 2021)*.

NETWORKED SYSTEMS

- **High-performance Emulation of Heterogeneous Systems using Adaptive Time Dilation.**
Hee Won Lee, Mihail L. Sichitiu, and David Thuente.
International Journal of High Performance Computing Applications, vol. 29, issue 2, May 2015.
- **Integrated Simulation and Emulation using Adaptive Time Dilation.**
Hee Won Lee, David Thuente, and Mihail L. Sichitiu.
 In *ACM 2nd SIGSIM Conference on Principles of Advanced Discrete Simulation (PADS 2014)*.
- **Network Link Emulation With Adaptive Time Dilation.**
Hee Won Lee, Mihail L. Sichitiu, and David Thuente.
Journal of Parallel and Distributed Computing, vol. 104, pp. 88–98, Jun 2017.
- **Accelerating Applications in the Fast-Moving Devices with Proactive Provisioning (Poster).**
 HyunJong Lee, Hee Won Lee, Moo-Ryong Ra, Yu Xiang, and Jason Flinn.
 In *Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services (MobiSys 2019)*.

PATENTS

Dynamic Provisioning of Storage in the Cloud.

Moo-ryong Ra and Hee Won Lee.
 US 10,530,703 B2 – Date of Patent: January 7, 2020.

[CV compiled on 2025-06-22]